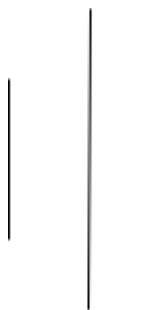


Medical Insurance Premium Predictor

“AI, Machine Learning, and NLP Powered Model for Predicting Insurance Expenses”



Submitted By:

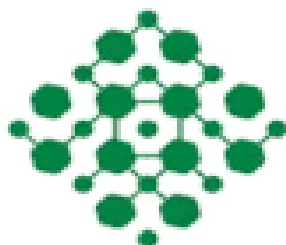
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1) Abstract:

This project focuses on predicting **medical insurance premiums** using both **structured** and **unstructured data**. Structured features include demographic and medical information such as **age, BMI(Body Mass Index), number of children, smoking status, and region**, while unstructured features include **claim descriptions, medical notes, and customer feedback**.

The project integrates **Machine Learning (ML)** and **Natural Language Processing (NLP)** techniques to build predictive models. Textual data is preprocessed and vectorized using **TF-IDF**, and numerical features are scaled to ensure uniformity. Multiple regression models, including **Linear Regression, Random Forest, Gradient Boosting, and XGBoost**, are trained and evaluated using metrics like **RMSE, MAE, and R² score**.

The best-performing model is deployed through a **Streamlit dashboard**, allowing users to input customer information and obtain **real-time premium predictions**. This approach demonstrates how AI can **leverage diverse data sources** to assist insurance companies in risk assessment, pricing strategies, and improving customer satisfaction.

Keywords: *Insurance Premium Prediction, Machine Learning, NLP, XGBoost, Random Forest, TF-IDF, Streamlit Dashboard*

2) Introduction:

The Medical Insurance Premium Prediction project aims to forecast individual insurance costs based on demographic, medical, and textual data. Insurance companies rely on accurate premium predictions to assess risk, optimize pricing, and maintain financial stability, while ensuring customers are charged fairly based on their health profile and lifestyle factors.

Traditional methods often use only structured data such as **age, BMI, number of children, smoking status, and region**. However, including **unstructured textual data**—like claim descriptions, medical notes, and customer feedback—can provide additional context and improve prediction accuracy.

This project combines **Machine Learning (ML)** and **Natural Language Processing (NLP)** techniques to handle both data types. Structured numerical and categorical data are preprocessed and scaled, while textual data is cleaned, tokenized, and vectorized using **TF-IDF**. Multiple regression models, including **Linear Regression, Random Forest, Gradient Boosting, and XGBoost**, are trained to predict insurance premiums.

The project also includes an interactive **Streamlit dashboard**, allowing users to input customer information and receive **real-time premium predictions**, making it practical for real-world insurance applications. By leveraging AI and ML, this project demonstrates how data-driven approaches can enhance **decision-making, pricing strategies, and customer satisfaction** in the insurance industry.

3) Aim of the Project:

The aim of this project is to **build an AI-powered system to accurately predict medical insurance premiums** by using both structured data (age, BMI, smoking status, etc.) and unstructured textual data (claim descriptions, medical notes, feedback).

This project focuses on:

- Improving premium prediction accuracy.
- Helping insurance companies assess risk and price fairly.
- Providing a simple, interactive platform for real-time predictions.

4) Problem Statement:

Insurance companies face the critical challenge of **accurately predicting individual medical insurance premiums**. Incorrect premium estimation can lead to either **financial losses for the company** or **overcharging customers**, which may impact customer satisfaction and trust.

Traditional premium calculations often rely solely on **structured data** such as age, BMI, number of children, smoking habits, and region. However, **ignoring unstructured textual data** like claim descriptions, medical notes, and customer feedback can result in **loss of valuable insights** regarding customer health risks and claim patterns.

The goal of this project is to develop a **data-driven predictive model** that combines both structured and unstructured data to forecast insurance premiums more accurately. By integrating **Machine Learning (ML)** and **Natural Language Processing (NLP)** techniques, the model aims to assist insurers in making **informed, fair, and efficient decisions** regarding premium pricing, ultimately improving risk assessment, customer satisfaction, and operational efficiency.

5) Objectives:

- To **predict individual medical insurance premiums** using both structured and unstructured data.
- To **preprocess and analyze structured data** (age, BMI, number of children, sex, smoker, region) for model training.
- To **process unstructured textual data** (claim descriptions, medical notes, customer feedback) using **Natural Language Processing (NLP)** techniques.
- To **implement and evaluate multiple Machine Learning models**, including Linear Regression, Random Forest, Gradient Boosting, and XGBoost, for premium prediction.
- To **combine structured and textual features** effectively for improving prediction accuracy.
- To **deploy an interactive dashboard** using Streamlit for real-time premium predictions.
- To provide **insights and recommendations** for insurance companies to optimize premium pricing and risk assessment.

6) Methodology:

The project follows a structured workflow to predict insurance premiums using both **structured** and **textual data**. The methodology can be divided into the following steps:

i. Data Collection

- The dataset (`insurance.csv`) contains **1338 records** and **7 features**:
 - **Numerical:** `age`, `bmi`, `children`, `expenses`
 - **Categorical:** `sex`, `smoker`, `region`
- Additional **textual features** such as `claim_description`, `medical_notes`, and `feedback` were generated to enhance model performance using NLP techniques.

ii. Importing Libraries and Dataset

In this project, several Python libraries were used for data analysis, NLP processing, model development, and deployment. Each library served a specific role in the workflow:

- **pandas, numpy** → For loading the dataset, handling missing values, and performing mathematical and statistical operations.
- **matplotlib, seaborn** → For Exploratory Data Analysis (EDA), including the creation of visualizations, correlation heatmaps, and distribution plots.
- **nltk, re, WordNetLemmatizer, stopwords** → For Natural Language Processing (NLP) tasks such as text cleaning, tokenization, stopwords removal, and lemmatization — especially useful for processing textual columns like claim descriptions, feedback, and medical notes.
- **LabelEncoder, StandardScaler** → For encoding categorical variables and scaling numerical features to prepare the data for model training.
- **train_test_split, cross_val_score, GridSearchCV** → For splitting the data into training and testing sets, validating model performance, and tuning hyperparameters for optimal results.
- **xgboost, sklearn.ensemble, sklearn.linear_model** → For building, training, and optimizing predictive machine learning models.
- **mean_squared_error, r2_score, mean_absolute_error** → For evaluating model accuracy and performance using regression metrics.

- streamlit, pickle → For deploying the trained model as an interactive web application and saving/loading model files for prediction.

iii. Data Preprocessing

- **Handling missing values:** Verified and confirmed there were no missing values in the dataset.
- **Outlier treatment:** Outliers in **bmi** and **expenses** were capped using **Interquartile Range (IQR)** and **Arbitrary Outlier Capping** to reduce skewness.
- **Skewness correction:** Log transformation applied to **expenses** to normalize highly skewed distribution.
- **Encoding categorical variables:**
 - **sex:** male → 0, female → 1
 - **smoker:** no → 0, yes → 1
 - **region:** northwest → 0, northeast → 1, southeast → 2, southwest → 3

iv. Text Preprocessing

- **Tokenization:** Breaking down text into individual words.
- **Lowercasing:** Standardizing all words to lowercase.
- **Stopword removal:** Removed common words using NLTK's stopwords.
- **Lemmatization:** Converted words to their root forms to reduce redundancy.
- **Vectorization:** Converted text data into numeric format using **TF-IDF** for model input.

v. Feature Engineering

- Structured numerical and categorical features were **scaled** using **StandardScaler**.
- Text features were **vectorized using TF-IDF** and combined with structured features using **sparse matrices (hstack)**.

vi. Model Development

- **Dataset split:** 80% training, 20% testing.

- **Model selection:** Multiple regression models were trained and evaluated:
 - Linear Regression, Lasso, Ridge, KNN Regressor
 - Decision Tree, Random Forest, Gradient Boosting, AdaBoost, XGBoost
- **Hyperparameter tuning:** Applied **RandomizedSearchCV** to optimize Random Forest and XGBoost models.
- **Model evaluation metrics:**
 - Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score

vii. Model Deployment

- Developed an **interactive Streamlit dashboard** where users can:
 - Enter structured inputs (age, BMI, children, sex, smoker, region)
 - Provide textual inputs (claim description, medical notes, feedback)
 - Obtain **real-time insurance premium predictions**
- The final model pipeline includes the **best performing ML model**, **scaler**, and **TF-IDF vectorizers** for seamless predictions.

viii. Insights and Use Cases

- The model helps insurance companies to:
 - Estimate **fair premiums** based on customer profile
 - Identify **high-risk individuals** using combined structured and textual features
 - **Enhance operational efficiency** and decision-making

7) Solution:

To solve this problem, the project implements an **AI-powered predictive system** that:

1. **Integrates Structured and Textual Data:** Combines demographic and medical features with unstructured textual information from claims, medical notes, and feedback.
2. **Applies Machine Learning and NLP Techniques:** Uses models like **Random Forest** and **XGBoost**, with TF-IDF vectorization for text, to predict premiums accurately.
3. **Provides Real-Time Predictions:** Deploys the model via a **Streamlit dashboard**, allowing users to input customer data and obtain instant, reliable premium estimates.
4. **Improves Decision-Making:** Helps insurance companies **price premiums fairly, assess risk efficiently, and make data-driven decisions**, addressing the limitations of traditional approaches

8) Results and Analysis:

The performance of various regression models was evaluated on the **insurance dataset** using **structured** and **textual features**. The models were assessed based on **Root Mean Squared Error (RMSE)**, **Mean Absolute Error (MAE)**, and **R² score**.

i. Model Performance Summary

Algorithm	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R ²	TestR ²
Random Forest	0.0703	0.1810	0.0337	0.0814	0.9936	0.9607
XGBoost	0.0120	0.1823	0.0081	0.0950	0.9998	0.9602
Gradient Boosting	0.1637	0.2073	0.0992	0.1148	0.9653	0.9485
Decision Tree	0.0000	0.2316	0.0000	0.1029	1.0000	0.9357
KNN	0.2386	0.3218	0.1581	0.2064	0.9263	0.8758
AdaBoost	0.3201	0.3266	0.2634	0.2662	0.8673	0.8721
Ridge	0.3519	0.3401	0.2834	0.2675	0.8397	0.8613
Linear Regression	0.3516	0.3403	0.2836	0.2686	0.8399	0.8611
Lasso	0.8788	0.9161	0.7127	0.7607	0.0000	-0.007

Observations:

- **Random Forest** achieved the **lowest Test RMSE (0.1810)** and highest Test R^2 (0.9607), making it the most reliable model for this dataset.
- **XGBoost** performed nearly as well, with Test RMSE 0.1823 and R^2 0.9602. Its extremely low train RMSE indicates strong fitting capability, though slightly overfit.
- **Gradient Boosting** also showed strong performance but slightly higher error than the top two models.
- **Linear models** (Ridge, Linear Regression, Lasso) had reasonable performance, but Lasso severely underperformed due to inappropriate regularization for this dataset.
- **Decision Tree** achieved perfect training scores (overfit) but generalization on the test set was lower.
- **KNN and AdaBoost** provided moderate accuracy but were not competitive with tree-based ensembles.

ii. Feature Analysis

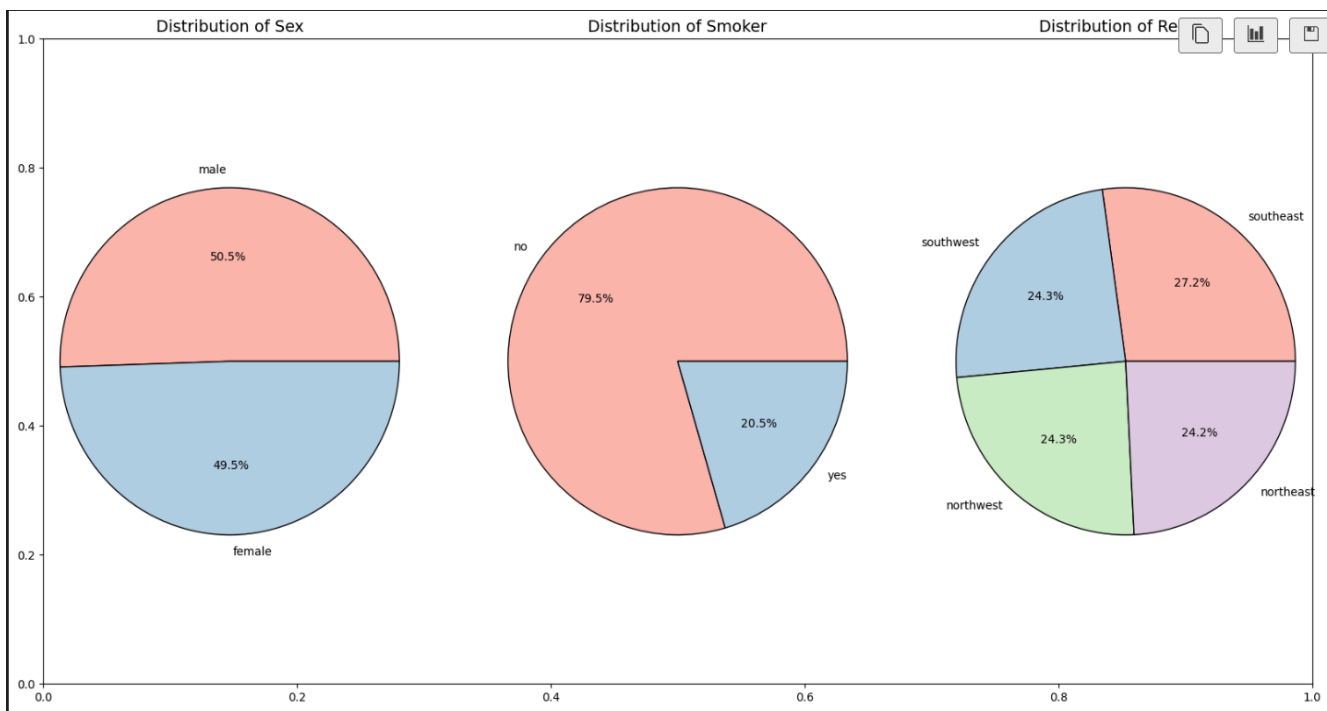
Structured Features

- **Smoker Status** and **BMI** are the strongest predictors of insurance expenses.
- **Age** and **Number of Children** have moderate influence.
- **Sex** and **Region** have minimal impact.

Textual Features (NLP)

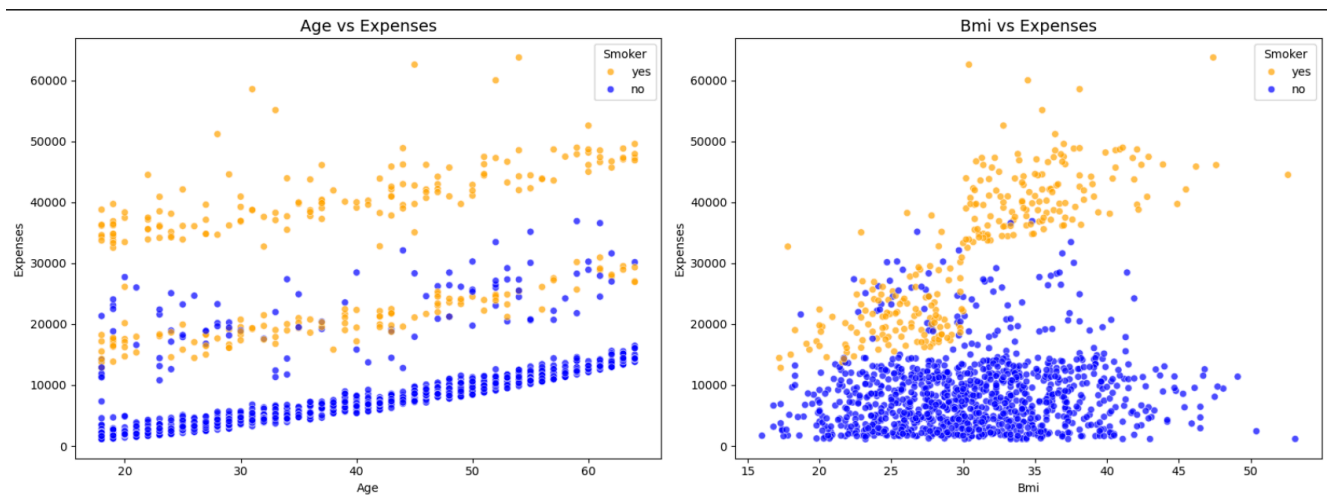
- **Claim Descriptions:** Keywords like *“accident”*, *“surgery”*, and *“critical illness”* strongly impact predicted premiums.
- **Medical Notes:** Chronic illnesses like diabetes and hypertension contribute to risk assessment.
- **Customer Feedback:** Minimal influence alone, but improves model accuracy when combined with other features.

9) Feature Visualization:



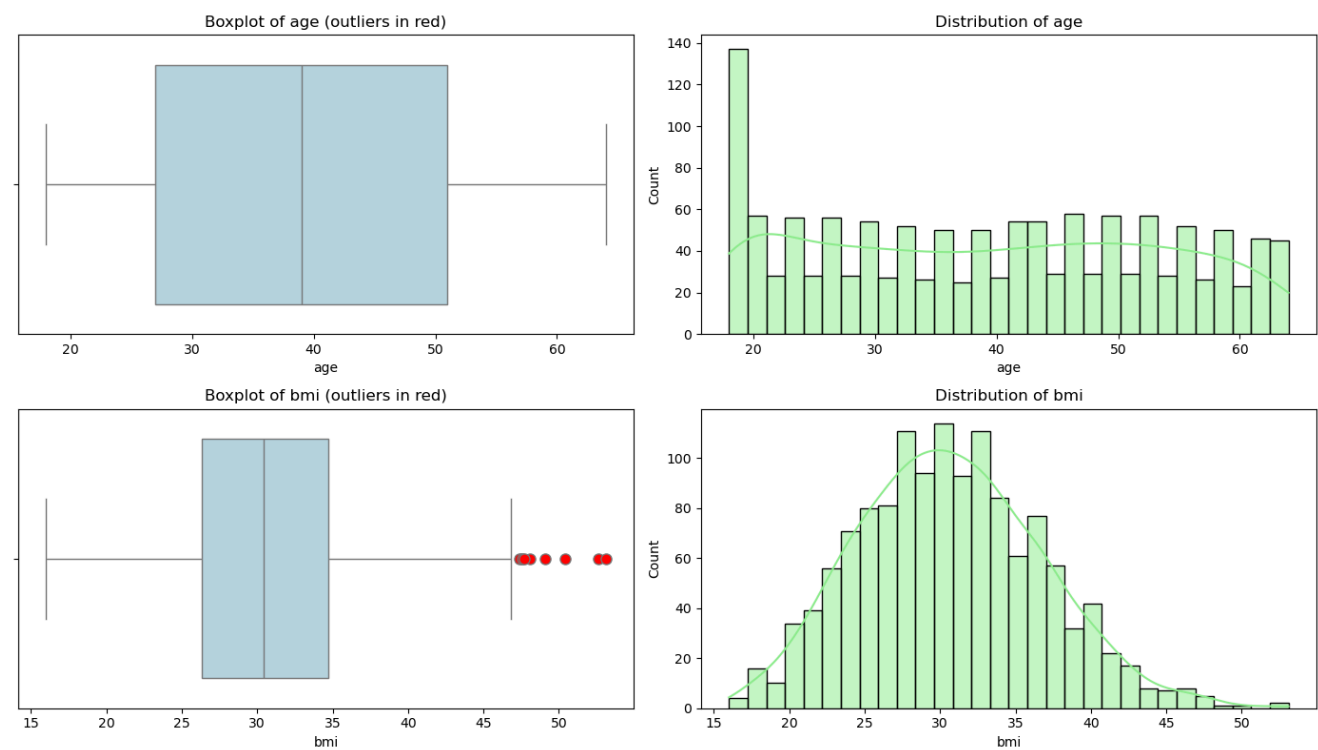
Pie chart for the sex, smoker, and region column

The data provided to us is equally distributed among the sex and the region columns but in the smoker column, we can observe a ratio of 80:20



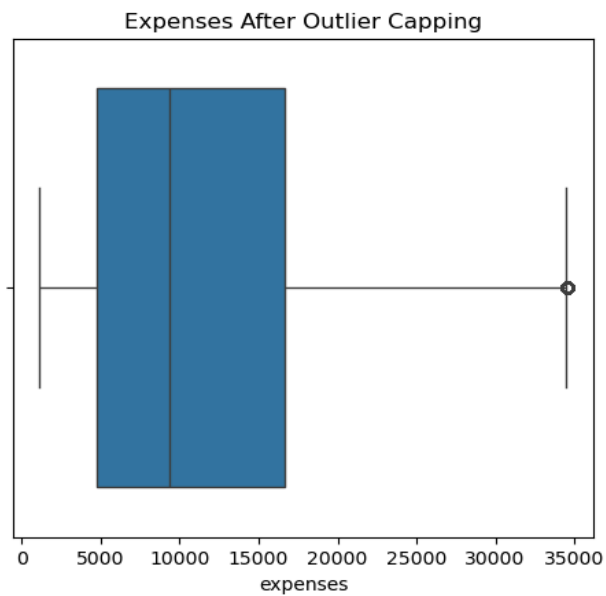
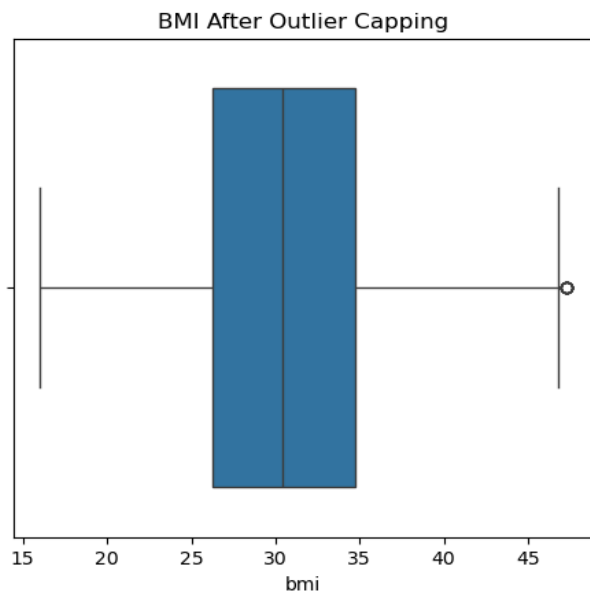
Scatter plot of the charges paid v/s age and BMI respectively

A clear distinction can be observed here between the charges that smokers have to pay. Also here as well we can observe that as the age of a person increases premium prices goes up as well



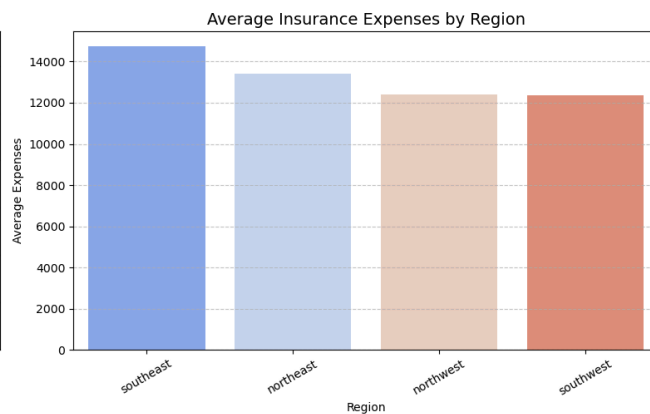
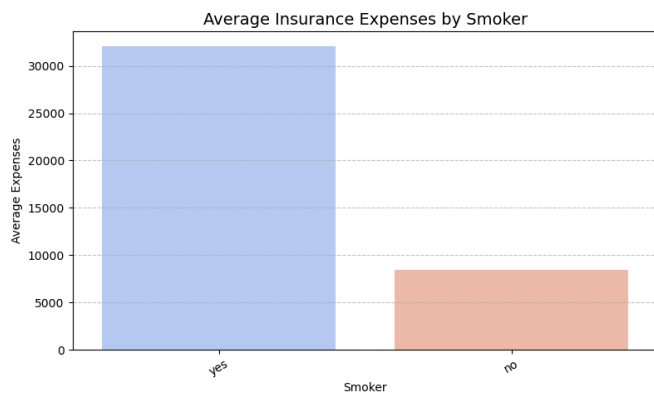
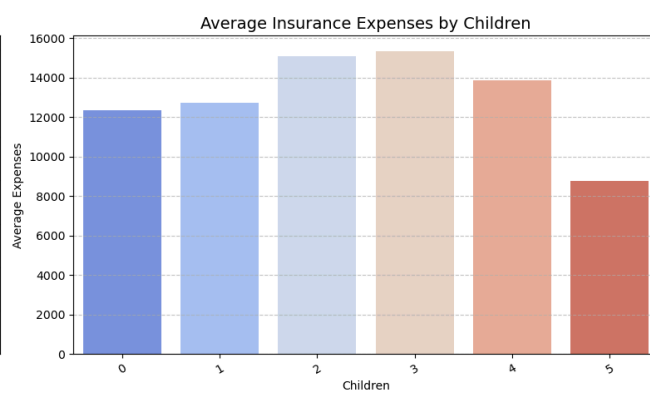
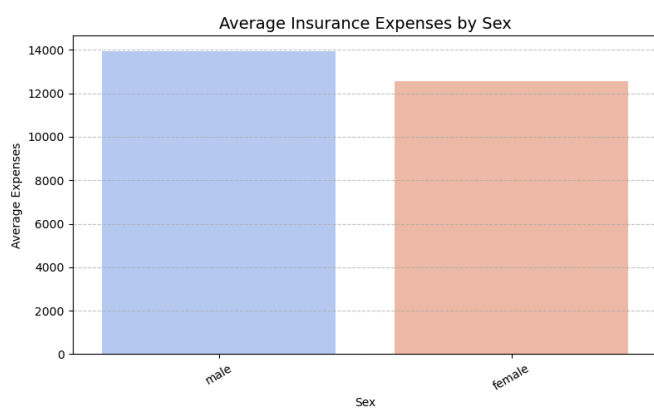
Box plot and histogram for bmi vs age

Due to the presence of outliers present in the bmi column we need to treat the outliers by replacing the values with mean as the bmi column consists of continuous data.



Box plot for bmi vs expenses

Now we successfully treated the outliers



Comparison between charges paid between different groups

- Charges are on the higher side for males as compared to females but the difference is not that much.
- Premium charged from the smoker is around thrice that which is charged from non-smokers.
- Charges are approximately the same in the given four regions

10) Conclusion:

- combining structured and NLP-based textual features improves prediction accuracy.
- XGBoost with tuned hyperparameters provided the most reliable predictions.
- The interactive Streamlit dashboard demonstrates practical application, allowing insurance companies to assess risk and calculate premiums efficiently.

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Customer Information

Age

30

BMI

25.00

Number of Children

0

Sex

Male

Smoker

No

Region

Northwest

Claim Description

Minor injury claim

Medical Notes

No chronic illness

Customer Feedback

Quick claim, no issues

Predict Insurance Expense

\$

Predicted Insurance Expense

\$3,394.22

Developed by Indra Jaiswal

Deploy

Dashboard